

MATH110BH Homework 4

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1 Problem 2

Lemma 1.1. Let $R = \mathbb{Z}[\sqrt{-5}]$. The ideal I generated by 2 and $1 + \sqrt{-5}$ is not principal.

Proof. Assume by contradiction that I is principal. Then, there's some $a \in R$ that divides both 2 and $1 + \sqrt{-5}$.

We use the norm argument. $N(2) = 4$ and $N(1 + \sqrt{-5}) = 6$, so $N(a) = 2$ so $a = 1$ or $a = 2$, both of which lead to a contradiction. $\square$

2 Problem 3

2 is not prime in $\mathbb{Z}[i]$ since $2 = (1 + i)(1 - i)$ and 2 doesn't divide either of them.

3 Problem 4

Lemma 3.1. Let p be a prime number such that $p \equiv 1 \pmod{4}$. Then, p is not prime in $\mathbb{Z}[i]$.

Proof. Since $p \equiv 1 \pmod{4}$, $\exists x \in \mathbb{Z} : x^2 \equiv -1 \pmod{p}$. Notice that $x \neq \pm 1$ since that would imply that p is even.

Let $\pi : \mathbb{Z}[i] \rightarrow (\mathbb{Z}/p\mathbb{Z}[i])$. Recall that $(\mathbb{Z}/p\mathbb{Z}[i])$ is isomorphic to $\mathbb{Z}[i]/(p)$. We'll denote values in the images of π by $\bar{x}$.

Notice $\bar{x} \pm i$ or isn't zero in $(\mathbb{Z}/p\mathbb{Z}[i])$.

However, $(\bar{x} - i)(\bar{x} + i) = 0$, so $\mathbb{Z}[i]/(p)$ isn't an integral domain. Thus, p isn't prime in $\mathbb{Z}[i]$. $\square$

4 Problem 6

Recall that $(a) \subseteq (b)$ if and only if $b \mid a$.

Since $\gcd(a, b)$ divides both a and b , it contains both principal ideals.

If (c) contains (a) and (b) , c divides both a and b , so c divides $\gcd(a, b)$. Then, the ideal generated by $\gcd(a, b)$ is contained in (c) .

5 Problem 7

Lemma 5.1. If $R[x]$ is a UFD, R is a UFD.

Proof. Recall from lecture that $(R[x])^\times = R^\times$.

Thus, R admits factorization by using the factorization in $R[x]$.

Let $r \in R$ be irreducible in $R[x]$. Then, r is irreducible in R . Thus, we can use the factorization in $R[x]$ to factor r , and this is unique using the factorization in $R[x]$. $\square$

6 Problem 8

Notice that $x^9 + y^9 = (x^3 + y^3)(x^6 - y^3x^3 + y^6)$.

Also notice that $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$.

Notice that $x + y$ doesn't divide $x^2 - xy + y^2$ since setting $y = -x$ makes this polynomial into $3x^2$ which isn't the zero polynomial. Similarly, $x + y$ doesn't divide $x^6 - y^3x^3 + y^6$.

Then, using Eisenstein's in the ring $\mathbb{C}[x, y][z]$ with $p = x + y$, we conclude the proof.

Notice that $x + y$ is prime in $\mathbb{C}[x, y]$ since quotienting out by $x + y$ produces a ring isomorphic to $\mathbb{C}[x]$ which is an integral domain.

7 Problem 10

Lemma 7.1. Let $R_1 \subset R_2 \subset \dots$ be a chain of countably many subrings of a ring R such that $R = \bigcup R_i$. Suppose that all the R_i are UFDs and any prime element in every R_i is prime in R_{i+1} . Then, R is a UFD.

Proof. Let's first show that R admits factorization into irreducibles. Let $r \in R$. Then, $r \in R_i$ for some $i \in \mathbb{N}$. Then, $r = c_1 \cdot \dots \cdot c_n$ where c_i is an irreducible element of R_i . Therefore, it suffices to show that c_i is irreducible in R . Let $c_i = uv$ for some $u, v \in R$. Then, $u, v \in R_j$ for some j by taking the maximum. Then, u or v is a unit in R_j . Since a unit in a smaller ring is a unit in the larger ring, u or v is a unit in R . Also notice that c_i doesn't become a unit in R since that would imply that it's a unit in some R_j , which contradicts our assumption that irreducibles stay irreducible.

Now, suppose by contradiction that factorization is not unique. Then, there is some $x \in R$ that has two different prime factorizations, i.e. $x = a_1 a_2 \dots a_n = b_1 b_2 \dots b_m$. Since a_i and b_j are all in R , they are all contained in some R_i and R_j . By taking the maximum of all of these indices, we get some R_i such that all a_i and b_i are in R_i and are prime in R_i . However, this contradicts the fact that R_i is a UFD. $\square$

Lemma 7.2. The polynomial ring $\mathbb{Z}[x_1, x_2, \dots]$ in countably many variables is a UFD but not a Noetherian ring.

Proof. Notice that $\mathbb{Z}[x_1, x_2, \dots] = \bigcup R_n$ where $R_n = \mathbb{Z}[x_1, x_2, \dots, x_n]$. We have $R_i \subset R_{i+1}$. Let f prime in R_i . Then, f is also prime in R_{i+1} since any polynomial involving x_{i+1} can't divide f . Thus, by the lemma above, $\mathbb{Z}[x_1, x_2, \dots]$ is a UFD.

Now, let I_n be the ideal generated by $(x_1, \dots, x_n)$. Clearly, $I_n \subset I_{n+1}$, but this increasing sequence of ideals never ends. Therefore, $\mathbb{Z}[x_1, x_2, \dots]$ is not Noetherian. $\square$